

Ricardo (Zhicong) Xie

MECHANICAL ENGINEERING | PROTOTYPING AND TESTING

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Portfolio: [zxie588-alt.github.io](https://github.com/zxie588-alt) | Expected graduation: July 2027

PROFILE

University of Auckland mechanical engineering master's student with hands-on experience in CAD, 3D printing, prototype assembly and soldering. Currently developing an energy-harvesting backpack under Lihua Tang. Interested in practical engineering roles involving product development, testing, data analysis and technical documentation.

EDUCATION

University of Auckland | Master of Engineering Studies - Mechanical Engineering

2026 - July 2027 (expected) | GPA: 6.0/9.0

- Relevant completed courses: Research Methods for Engineers (A), Composite Materials (B), Advanced Computational Fluid Dynamics (A-).

Huazhong Agricultural University | Bachelor of Engineering

Mechanical Engineering and Mechatronics | 2021 - 2025

RELEVANT PROJECT EXPERIENCE

Energy-Harvesting Backpack | University of Auckland | July 2026 - present

Supervisor: Lihua Tang

- Built the suspended-load platform shown in the KINELoad presentation: twin guides, elastic suspension and rack-and-pinion transmission integrated with the research group's existing generator. Used CAD, 3D printing, mechanical assembly and soldering.
- Participated in doctoral-researcher-led bench testing and demonstrations of storage-assisted phone charging and pulse-oximeter operation. The generator and team test results are not claimed as independent inventions or measurements.
- Identified that a 5 mm hole in a printed part would not accept an M5 bolt. Learned to account for assembly clearance and manufacturing variation rather than relying on nominal dimensions alone.

Dynamic Livestock Weighing System | STM32, HX711 and load cells

- Developed load-cell data acquisition, embedded logic and serial/Bluetooth transfer to a PC; tested signal stability and documented the sensing system.
- Registered as Computer Software Copyright No. 2024SR0141888.

Composite Formula SAE Front Wing Analysis | Abaqus

- Analysed a composite shell model under multiple loading conditions; compared ply-level stresses and documented assumptions, model limitations and the governing load case.

PRACTICAL AND TECHNICAL SKILLS

- **Hands-on:** 3D printing, prototype assembly, soldering, mechanical fit checks, sensors and data logging.
- **Analysis and documentation:** Excel, engineering calculations, interpreting results and documenting technical work; MATLAB/Simulink and basic Python.
- **Engineering software:** SolidWorks, AutoCAD/DXF, Abaqus and ANSYS Fluent/CFX.

COMMUNICATION AND LEADERSHIP

- **Photography Director | Chinese Postgraduate Society (CNPg), New Zealand** - current committee role in photography.
- Former undergraduate student-news department head and photographer; experience coordinating work and communicating with different people.
- Discuss project progress and practical questions with a university supervisor; comfortable asking for guidance and learning from feedback.